

17

Dividends and Payout Policy

ON SEPTEMBER 19, 2019, Microsoft announced a broad plan to reward stockholders for the recent success of the firm's business. Under the plan, Microsoft would (1) boost its annual dividend by 11 percent, from \$.45 per share to \$.51 per share, and (2) repurchase about \$40 billion of Microsoft's common stock. Investors cheered, bidding up the stock price by as much as 3 percent on the day of the announcement, on a day when the stock market as a whole fell. Why were investors so pleased? To find out, this chapter explores these actions and their implications for shareholders.

LEARNING OBJECTIVES

After studying this chapter, you should be able to:

- | | |
|--|---|
| L01 Define dividend types and how dividends are paid. | L03 Describe the difference between cash and stock dividends. |
| L02 Explain the issues surrounding dividend policy decisions. | L04 Explain why share repurchases are an alternative to dividends. |

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Dividend policy is an important subject in corporate finance, and dividends are a major cash outlay for many corporations. For example, S&P 500 companies paid about \$485 billion in dividends in 2019. In fact, dividends globally were \$1.4 trillion. Microsoft and Apple were the biggest payers. How big? Both Microsoft and Apple paid out about \$14 billion each in dividends during 2019. In contrast, about 15 percent of the companies in the S&P 500 paid no dividend at all.

Of course, the 2020 economic slowdown caused by COVID-19 negatively affected dividends. For example, Shell announced it would slice its quarterly dividend from \$.47 to \$.16, the company's first dividend cut since World War II. The decision freed up \$10 billion in cash for Shell. Other companies, including Southwest Airlines, Weyerhaeuser, and Walt Disney, suspended dividends entirely. Overall, worldwide dividends were expected to decline by 15 to 35 percent in 2020.

At first glance, it may seem obvious that a firm would always want to give as much as possible back to its shareholders by paying dividends. It might seem equally obvious that a firm could always invest the money for its shareholders instead of paying it out. The heart of the dividend policy question is this: Should the firm pay out money to its shareholders, or should the firm take that money and invest it for its shareholders?



In this chapter, we will cover a variety of topics related to dividends and corporate payout policies. We first discuss the various types of cash dividends and how they are paid. We ask whether dividend policy matters, and we consider arguments in favor of both high and low dividend payouts. Next, we examine stock repurchases, which have become an important alternative to cash dividends. We then bring together several decades of research on dividends and corporate payouts to describe the key trade-offs involved in establishing a payout policy. We conclude the chapter by discussing stock splits and stock dividends.

Cash Dividends and Dividend Payment

The term **dividend** usually refers to cash paid out of earnings. If a payment is made from sources other than current or accumulated retained earnings, the term **distribution**, rather than *dividend*, is used. It is acceptable to refer to a distribution from earnings as a dividend and a distribution from capital as a liquidating dividend. More generally, any direct payment by the corporation to the shareholders may be considered a dividend or a part of dividend policy.

Dividends come in several different forms. The basic types of cash dividends are these:

1. Regular cash dividends.
2. Extra dividends.
3. Special dividends.
4. Liquidating dividends.

Later in the chapter, we discuss dividends paid in stock instead of cash. We also consider another alternative to cash dividends: stock repurchase.

CASH DIVIDENDS

The most common type of dividend is a cash dividend. Commonly, public companies pay **regular cash dividends** four times per year. As the name suggests, these are cash payments made directly to shareholders, and they are made in the regular course of business. In other words, management sees nothing unusual about the dividend and no reason why it won't be continued.

Sometimes firms will pay a regular cash dividend and an *extra cash dividend*. By calling part of the payment “extra,” management is indicating that the “extra” part may or may not be repeated in the future. A *special dividend* is similar, but the name usually indicates that this dividend is viewed as a truly unusual or one-time event and won't be repeated. For example, in December 2004, Microsoft paid a special dividend of \$3 per share. The total payout of \$32 billion was the largest one-time corporate dividend in history. Founder Bill Gates received about \$3 *billion*, which he pledged to donate to charity. To give you another idea of the size of the special dividend, consider that in December, when the dividend was sent to investors, personal income in the United States rose 3.7 percent. Without the dividend, personal income rose only .3 percent, so the dividend payment accounted for about 3 percent of all personal income in the United States for the month! Finally, the payment of a *liquidating dividend* usually means that some or all of the business has been liquidated—that is, sold off.

However it is labeled, a cash dividend payment reduces corporate cash and retained earnings, except in the case of a liquidating dividend (which may reduce paid-in capital).

STANDARD METHOD OF CASH DIVIDEND PAYMENT

The decision to pay a dividend rests in the hands of the board of directors of the corporation. When a dividend has been declared, it becomes a debt of the firm and cannot easily be rescinded. Sometime after it has been declared, a dividend is distributed to all shareholders as of some specific date.

17.1

dividend

A payment made out of a firm's earnings to its owners, in the form of either cash or stock.

distribution

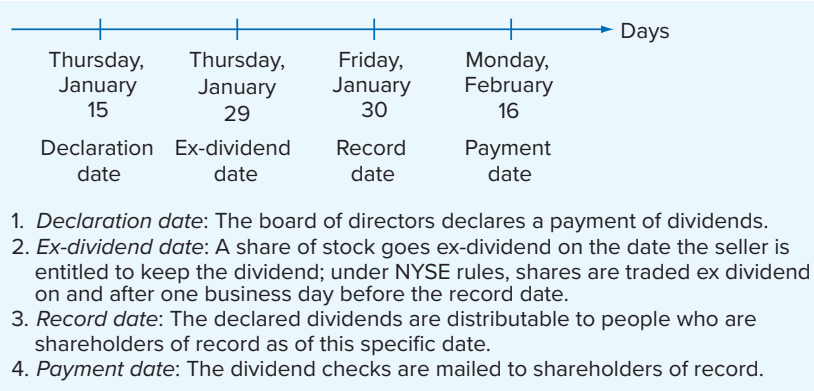
A payment made by a firm to its owners from sources other than current or accumulated retained earnings.

regular cash dividend

A cash payment made by a firm to its owners in the normal course of business, usually paid four times per year.



For a list of companies that are paying dividends today, go to www.thestreet.com/dividends.

**FIGURE 17.1****Example of
Procedure for
Dividend Payment**

Commonly, the amount of the cash dividend is expressed in terms of dollars per share (*dividends per share*). As we have seen in other chapters, it is also expressed as a percentage of the market price (the *dividend yield*) or as a percentage of net income or earnings per share (the *dividend payout*).

DIVIDEND PAYMENT: A CHRONOLOGY

The mechanics of a cash dividend payment can be illustrated by the example in Figure 17.1 and the following description:

declaration date

The date on which the board of directors passes a resolution to pay a dividend.

ex-dividend date

The date one business day before the date of record, establishing those individuals entitled to a dividend.

date of record

The date by which a holder must be on record to be designated to receive a dividend.

date of payment

The date on which the dividend checks are mailed.

1. **Declaration date:** On January 15, the board of directors passes a resolution to pay a dividend of \$1 per share on February 16 to all holders of record as of January 30.
2. **Ex-dividend date:** To make sure that dividend checks go to the right people, brokerage firms and stock exchanges establish an ex-dividend date. This date is one business day before the date of record (discussed next). If you buy the stock before this date, you are entitled to the dividend. If you buy on this date or after, the previous owner will get the dividend.

In Figure 17.1, Thursday, January 29, is the ex-dividend date. Before this date, the stock is said to trade “with dividend” or “cum dividend.” Afterward, the stock trades “ex dividend.”

The ex-dividend date convention removes any ambiguity about who is entitled to the dividend. Because the dividend is valuable, the stock price will be affected when the stock goes “ex.” We examine this effect in a moment.

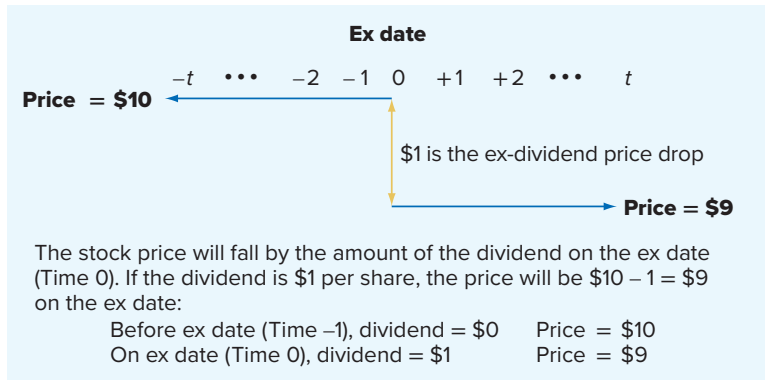
3. **Date of record:** Based on its records, the corporation prepares a list on January 30 of all individuals believed to be stockholders. These are the *holders of record*, and January 30 is the date of record (or record date). The word *believed* is important here. If you buy the stock just before this date, the corporation’s records may not reflect that fact because of mailing or other delays. Without some modification, some of the dividend checks will get mailed to the wrong people. This is the reason for the ex-dividend date convention.
4. **Date of payment:** The dividend checks are mailed on February 16.

MORE ABOUT THE EX-DIVIDEND DATE

The ex-dividend date is important and is a common source of confusion. We examine what happens to the stock when it goes ex, meaning that the ex-dividend date arrives.

**FIGURE 17.2**

**Price Behavior
around the
Ex-Dividend Date for
a \$1 Cash Dividend**



To illustrate, suppose we have a stock that sells for \$10 per share. The board of directors declares a dividend of \$1 per share, and the record date is set to be Monday, June 11. Based on our previous discussion, we know that the ex date will be one business (not calendar) day earlier, on Friday, June 8.

If you buy the stock on Thursday, June 7, just as the market closes, you'll get the \$1 dividend because the stock is trading cum dividend. If you wait and buy it just as the market opens on Friday, you won't get the \$1 dividend. What happens to the value of the stock overnight?

If you think about it, you will see that the stock is worth about \$1 less on Friday morning, so its price will drop by this amount between close of business on Thursday and the Friday opening. In general, we expect that the value of a share of stock will go down by about the dividend amount when the stock goes ex dividend. The key word here is *about*. Because dividends are taxed, the actual price drop might be closer to some measure of the aftertax value of the dividend. Determining this value is complicated because of the different tax rates and tax rules that apply for different buyers.

The series of events described here is illustrated in Figure 17.2.

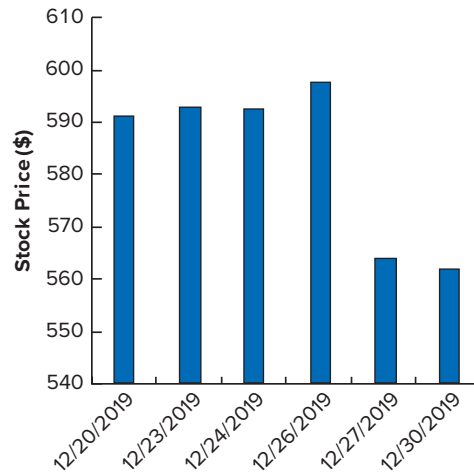
"Ex" Marks the Day

EXAMPLE 17.1

The board of directors of Divided Airlines has declared a dividend of \$2.50 per share payable on Tuesday, May 30, to shareholders of record as of Tuesday, May 9. Cal Icon buys 100 shares of Divided on Tuesday, May 2, for \$150 per share. What is the ex date? Describe the events that will occur with regard to the cash dividend and the stock price.

The ex date is one business day before the date of record, Tuesday, May 9; so the stock will go ex on Monday, May 8. Cal buys the stock on Tuesday, May 2, so Cal purchases the stock cum dividend. In other words, Cal will get $\$2.50 \times 100 = \250 in dividends. The check will be mailed on Tuesday, May 30. Just before the stock does go ex on Monday, its value will drop overnight by about \$2.50 per share.

As an example of the price drop on the ex-dividend date, we examine the large dividend paid by aerospace company TransDigm Group in December 2019. The dividend was \$32.50 per share at a time when the stock price was around \$600, so the dividend was about 5 percent of the total stock price, a truly special dividend. The stock went ex dividend on December 27, 2019. The stock price chart here shows the change in TransDigm during December of that year.



The stock closed at \$597.78 on December 26 and opened at \$568.00 on December 27—a drop of \$29.78. With a 20 percent tax rate on dividends, we would have expected a drop of about \$26, so the actual price dropped more than we would have expected. We discuss dividends and taxes in more detail in a subsequent section.

Concept Questions

- 17.1a** What are the different types of cash dividends?
- 17.1b** What are the mechanics of the cash dividend payment?
- 17.1c** How should the price of a stock change when it goes ex dividend?

17.2 Does Dividend Policy Matter?

To decide whether or not dividend policy matters, we first have to define what we mean by dividend *policy*. All other things being the same, of course dividends matter. Dividends are paid in cash, and cash is something that everybody likes. The question we will be discussing here is whether the firm should pay out cash now or invest the cash and pay it out later. Dividend policy is the time pattern of dividend payout. In particular, should the firm pay out a large percentage of its earnings now or a small (or even zero) percentage? This is the dividend policy question.

AN ILLUSTRATION OF THE IRRELEVANCE OF DIVIDEND POLICY

A powerful argument can be made that dividend policy does not matter. We illustrate this by considering the case of Wharton Corporation. Wharton is an all-equity firm that has existed for 10 years. The current financial managers plan to dissolve the firm in two years. The total cash flows the firm will generate, including the proceeds from liquidation, will be \$10,000 in each of the next two years.

Current Policy: Dividends Set Equal to Cash Flow At the present time, dividends at each date are set equal to the cash flow of \$10,000. There are 100 shares outstanding, so the dividend per share is \$100. In Chapter 6, we showed that the value of the

stock is equal to the present value of the future dividends. Assuming a 10 percent required return, the value of a share of stock today, P_0 , is:

$$\begin{aligned} P_0 &= \frac{D_1}{(1+R)^1} + \frac{D_2}{(1+R)^2} \\ &= \frac{\$100}{1.10} + \frac{\$100}{1.10^2} = \$173.55 \end{aligned}$$

The firm as a whole is worth $100 \times \$173.55 = \$17,355$.

Several members of the board of Wharton have expressed dissatisfaction with the current dividend policy and have asked you to analyze an alternative policy.

Alternative Policy: Initial Dividend Greater than Cash Flow Another possible policy is for the firm to pay a dividend of \$110 per share on the first date (Date 1), which is, of course, a total dividend of \$11,000. Because the cash flow is only \$10,000, an extra \$1,000 must somehow be raised. One way to do this is to issue \$1,000 worth of bonds or stock at Date 1. Assume that stock is issued. The new stockholders will desire enough cash flow at Date 2 so that they earn the required 10 percent return on their Date 1 investment.¹

What is the value of the firm with this new dividend policy? The new stockholders invest \$1,000. They require a 10 percent return, so they will demand $\$1,000 \times 1.10 = \$1,100$ of the Date 2 cash flow, leaving only \$8,900 to the old stockholders. The dividends to the old stockholders will be as follows:

	Date 1	Date 2
Aggregate dividends to old stockholders	\$11,000	\$8,900
Dividends per share	110	89

The present value of the dividends per share is:

$$P_0 = \frac{\$110}{1.10} + \frac{\$89}{1.10^2} = \$173.55$$

This is the same value we had before.

The value of the stock is unaffected by this switch in dividend policy even though we have to sell some new stock to finance the new dividend. In fact, no matter what pattern of dividend payout the firm chooses, the value of the stock will always be the same in this example. In other words, for the Wharton Corporation, dividend policy makes no difference. The reason is simple: Any increase in a dividend at some point in time is exactly offset by a decrease at some other time; so the net effect, once we account for time value, is zero.

HOMEMADE DIVIDENDS

There is an alternative and perhaps more intuitively appealing explanation of why dividend policy doesn't matter in our example. Suppose individual Investor X prefers dividends per share of \$100 at both Dates 1 and 2. Would she be disappointed if informed that the firm's management was adopting the alternative dividend policy (dividends of \$110 and \$89 on the two dates, respectively)? Not necessarily: She could easily reinvest the \$10 of

¹The same results would occur after an issue of bonds, though the arguments would be less easily presented.

unneeded funds received on Date 1 by buying more Wharton stock. At 10 percent, this investment would grow to \$11 by Date 2. Investor X would receive her desired net cash flow of $\$110 - 10 = \100 at Date 1 and $\$89 + 11 = \100 at Date 2.

Conversely, imagine that Investor Z, preferring \$110 of cash flow at Date 1 and \$89 of cash flow at Date 2, finds that management will pay dividends of \$100 at both Dates 1 and 2. This investor can sell \$10 worth of stock to boost his total cash at Date 1 to \$110. Because this investment returns 10 percent, Investor Z gives up \$11 at Date 2 ($= \10×1.1), leaving him with $\$100 - 11 = \89 .

Our two investors are able to transform the corporation's dividend policy into a different policy by buying or selling on their own. The result is that investors are able to create a **homemade dividend policy**. This means that dissatisfied stockholders can alter the firm's dividend policy to suit themselves. As a result, there is no particular advantage to any one dividend policy the firm might choose.

Many corporations actually assist their stockholders in creating homemade dividend policies by offering *automatic dividend reinvestment plans* (ADRs or DRIPs). McDonald's, Walmart, and Procter & Gamble, for example, have set up such plans. As the name suggests, with such a plan, stockholders have the option of automatically reinvesting some or all of their cash dividend in shares of stock. In some cases, they actually receive a discount on the stock, which makes such plans very attractive.

homemade dividend policy

The tailored dividend policy created by individual investors who undo corporate dividend policy by reinvesting dividends or selling shares of stock.



For more information on DRIPs, check out the Motley Fool at www.fool.com.

A TEST

Our discussion to this point can be summarized by considering the following true-false test questions:

1. True or false: Dividends are irrelevant.
2. True or false: Dividend policy is irrelevant.

The first statement is surely false, and the reason follows from common sense. Clearly, investors prefer higher dividends to lower dividends at any single date if the dividend level is held constant at every other date. To be more precise regarding the first question, if the dividend per share at a given date is raised while the dividend per share at every other date is held constant, the stock price will rise. The reason is that the present value of the future dividends must go up if this occurs. This action can be accomplished by management decisions that improve productivity, increase tax savings, strengthen product marketing, or otherwise improve cash flow.

The second statement is true, at least in the simple case we have been examining. Dividend policy by itself cannot raise the dividend at one date while keeping it the same at all other dates. Rather, dividend policy merely establishes the trade-off between dividends at one date and dividends at another date. Once we allow for time value, the present value of the dividend stream is unchanged. In this simple world, dividend policy does not matter because managers choosing either to raise or to lower the current dividend do not affect the current value of their firm. However, we have ignored several real-world factors that might lead us to change our minds; we pursue some of these in subsequent sections.

Concept Questions

- 17.2a** How can an investor create a homemade dividend?
17.2b Are dividends irrelevant?

Real-World Factors Favoring a Low Dividend Payout

17.3

The example we used to illustrate the irrelevance of dividend policy ignored taxes and flotation costs. In this section, we will see that these factors might lead us to prefer a low dividend payout.

TAXES

U.S. tax laws are complex, and they affect dividend policy in a number of ways. The key tax feature has to do with the taxation of dividend income and capital gains. Years ago, dividends received were taxed as ordinary income. Capital gains were taxed at somewhat lower rates, and the tax on a capital gain is deferred until the stock is sold. This second aspect of capital gains taxation makes the effective tax rate much lower because the present value of the tax is less.²

Tax law changes over time have led to a renewed interest in the effect of taxes on corporate dividend policies. Beginning in 2003, tax rates on dividends and long-term capital gains were lowered from a maximum in the 35–39 percent range to 15 percent, giving corporations a much larger tax incentive to pay dividends. In 2020, the tax rate on dividends was 0 percent, 15 percent, or 20 percent, depending on the individual's marginal tax rate. Even so, the fact that the capital gains tax can be deferred still means that the effective tax rate on dividends will be higher for many investors.

FLOTATION COSTS

In our example illustrating that dividend policy doesn't matter, we saw that the firm could sell some new stock if necessary to pay a dividend. As we mentioned in Chapter 15, selling new stock can be very expensive. If we include flotation costs in our argument, then we will find that the value of the stock decreases if we sell new stock.

More generally, imagine two firms identical in every way except that one pays out a greater percentage of its cash flow in the form of dividends. Because the other firm plows back more cash, its equity grows faster. If these two firms are to remain identical, then the one with the higher payout will have to periodically sell some stock to catch up. Because this is expensive, a firm might be inclined to have a low payout.

DIVIDEND RESTRICTIONS

In some cases, a corporation may face restrictions on its ability to pay dividends. For example, as we discussed in Chapter 7, a common feature of a bond indenture is a covenant prohibiting dividend payments above some level. Also, a corporation may be prohibited by state law from paying dividends if the dividend amount exceeds the firm's retained earnings.

²In fact, capital gains taxes can sometimes be avoided altogether. Although we do not recommend this particular tax avoidance strategy, the capital gains tax may be avoided by dying. Your heirs are not considered to have a capital gain, so the tax liability dies when you do. In this instance, you *can* take it with you.

Concept Questions**17.3a** What are the tax benefits of low dividends?**17.3b** Why do flotation costs favor a low payout?

17.4 Real-World Factors Favoring a High Dividend Payout

In this section, we consider reasons why a firm might pay its shareholders higher dividends even if it means the firm must issue more shares of stock to finance the dividend payments.

In a classic textbook, Benjamin Graham, David Dodd, and Sidney Cottle argue that firms should generally have high dividend payouts because:

1. “The discounted value of near dividends is higher than the present worth of distant dividends.”
2. Between “two companies with the same general earning power and same general position in an industry, the one paying the larger dividend will almost always sell at a higher price.”³

Two additional factors favoring a high dividend payout have also been mentioned frequently by proponents of this view: the desire for current income and the resolution of uncertainty.

DESIRE FOR CURRENT INCOME

It has been argued that many individuals desire current income. The classic example is the group of retired people and others living on a fixed income (the proverbial widows and orphans). It is argued that this group is willing to pay a premium to get a higher dividend yield. If this is true, then it lends support to the second claim made by Graham, Dodd, and Cottle.

It is easy to see that this argument is irrelevant in our simple case. An individual preferring high current cash flow but holding low-dividend securities can easily sell off shares to provide the necessary funds. Similarly, an individual desiring a low current cash flow but holding high-dividend securities can just reinvest the dividend. This is just our homemade dividend argument again. In a world of no transaction costs, a policy of high current dividends would be of no value to the stockholder.

The current income argument may have relevance in the real world. Here the sale of low-dividend stocks would involve brokerage fees and other transaction costs. These direct cash expenses could be avoided by an investment in high-dividend securities. In addition, the expenditure of the stockholder’s own time in selling securities and the natural (though not necessarily rational) fear of consuming out of principal might further lead many investors to buy high-dividend securities.

Even so, to put this argument in perspective, remember that financial intermediaries such as mutual funds can (and do) perform these “repackaging” transactions for individuals at very low cost. Such intermediaries could buy low-dividend stocks and, through a controlled policy of realizing gains, they could pay their investors at a higher rate.

³B. Graham, D. Dodd, and S. Cottle, *Security Analysis* (New York: McGraw-Hill, 1962).

TAX AND OTHER BENEFITS FROM HIGH DIVIDENDS

Earlier, we saw that dividends were taxed unfavorably for individual investors (at least until very recently). This fact is a powerful argument for a low payout. However, there are a number of other investors who do not receive unfavorable tax treatment from holding high-dividend yield, rather than low-dividend yield, securities.

Corporate Investors A significant tax break on dividends occurs when a corporation owns stock in another corporation. A corporate stockholder receiving either common or preferred dividends is granted a 50 percent (or more) dividend exclusion (reduced from 70 percent or more by the 2017 Tax Cuts and Jobs Act). Because the 50 percent exclusion does not apply to capital gains, this group is taxed unfavorably on capital gains.

As a result of the dividend exclusion, high-dividend, low-capital gains stocks may be more appropriate for corporations to hold. As we discuss elsewhere, this is why corporations hold a substantial percentage of the outstanding preferred stock in the economy. This tax advantage of dividends also leads some corporations to hold high-yielding stocks instead of long-term bonds because there is no similar tax exclusion of interest payments to corporate bondholders.

Tax-Exempt Investors We have pointed out both the tax advantages and the tax disadvantages of a low dividend payout. Of course, this discussion is irrelevant to those in zero tax brackets. This group includes some of the largest investors in the economy, such as pension funds, endowment funds, and trust funds.

CONCLUSION

Overall, individual investors (for whatever reason) may have a desire for current income and may be willing to pay the dividend tax. In addition, some very large investors such as corporations and tax-free institutions may have a very strong preference for high dividend payouts.

Concept Questions

- 17.4a** Why might some individual investors favor a high dividend payout?
- 17.4b** Why might some nonindividual investors prefer a high dividend payout?

A Resolution of Real-World Factors?

17.5

In the previous sections, we presented some factors that favor a low-dividend policy and others that favor a high-dividend policy. In this section, we discuss two important concepts related to dividends and dividend policy: the information content of dividends and the clientele effect. The first topic illustrates both the importance of dividends in general and the importance of distinguishing between dividends and dividend policy. The second topic suggests that, despite the many real-world considerations we have discussed, the dividend payout ratio may not be as important as we originally imagined.

INFORMATION CONTENT OF DIVIDENDS

To begin, we quickly review some of our earlier discussion. Previously, we examined three different positions on dividends:

1. Based on the homemade dividend argument, dividend policy is irrelevant.
2. Because of tax effects for individual investors and new issue costs, a low-dividend policy is best.
3. Because of the desire for current income and related factors, a high-dividend policy is best.

If you wanted to decide which of these positions is the right one, an obvious way to get started would be to look at what happens to stock prices when companies announce dividend changes. You would find with some consistency that stock prices rise when the current dividend is unexpectedly increased, and they generally fall when the dividend is unexpectedly decreased. What does this imply about any of the three positions just stated?

At first glance, the behavior we describe seems consistent with the third position and inconsistent with the other two. In fact, many writers have argued this position. If stock prices rise in response to dividend increases and fall in response to dividend decreases, then isn't the market saying that it approves of higher dividends?

Other authors have pointed out that this observation doesn't really tell us much about dividend policy. Everyone agrees that dividends are important, all other things being equal. Companies cut dividends only with great reluctance. Thus, a dividend cut is often a signal that the firm is in trouble.

More to the point, a dividend cut is usually not a voluntary, planned change in dividend policy. Instead, it usually signals that management does not think that the current dividend policy can be maintained. As a result, expectations of future dividends should generally be revised downward. The present value of expected future dividends falls, and so does the stock price.

In this case, the stock price declines following a dividend cut because future dividends are generally expected to be lower, not because the firm has changed the percentage of its earnings it will pay out in the form of dividends.

For example, on December 19, 2019, U.S. Steel announced that it was cutting its dividend by 80 percent, from \$.20 per year to \$.04 per year. The company stated that the drastic fall in steel prices was to blame. Shareholders were in for a big shock. On a typical day, about 14 million shares of the company stock traded hands. Following the announcement, about 43 million shares were traded and the stock lost about 11 percent of its value. And on the day Shell announced the dividend cut we discussed earlier in this chapter, its stock price fell about 13 percent.

Of course, the phenomenon of a stock price decrease in the face of a dividend cut is not restricted to the United States. In May 2019, Nissan announced that its annual dividend would be ¥40, well below the previous year's dividend of ¥57. In response, the stock dropped by about 7 percent.

In a similar vein, an unexpected increase in the dividend signals good news. Management will raise the dividend only when future earnings, cash flow, and general prospects are expected to rise to such an extent that the dividend will not have to be cut later. A dividend increase is management's signal to the market that the firm is expected to do well. The stock price reacts favorably because expectations of future dividends are revised upward, not because the firm has increased its payout.

In both of these cases, the stock price reacts to the dividend change. The reaction can be attributed to changes in the expected amount of future dividends, not necessarily a change in dividend payout policy. This reaction is called the **information content effect** of the dividend. The fact that dividend changes convey information about the firm to the market makes it difficult to interpret the effect of the dividend policy of the firm.

information content effect

The market's reaction to a change in corporate dividend payout.

THE CLIENTELE EFFECT

In our earlier discussion, we saw that some groups (wealthy individuals, for example) have an incentive to pursue low-payout (or zero-payout) stocks. Other groups (corporations, for example) have an incentive to pursue high-payout stocks. Companies with high payouts will attract one group, and low-payout companies will attract another.

These different groups are called *clienteles*, and what we have described is a **clientele effect**. The clientele effect argument states that different groups of investors desire different levels of dividends. When a firm chooses a particular dividend policy, the only effect is to attract a particular clientele. If a firm changes its dividend policy, then it attracts a different clientele.

What we are left with is a simple supply and demand argument. Suppose 40 percent of all investors prefer high dividends, but only 20 percent of the firms pay high dividends. Here the high-dividend firms will be in short supply; their stock prices will rise. Consequently, low-dividend firms will find it advantageous to switch policies until 40 percent of all firms have high payouts. At this point, the *dividend market* is in equilibrium. Further changes in dividend policy are pointless because all of the clienteles are satisfied. The dividend policy for any individual firm is now irrelevant.

To see if you understand the clientele effect, consider the following statement: In spite of the theoretical argument that dividend policy is irrelevant or that firms should not pay dividends, many investors like high dividends; because of this fact, a firm can boost its share price by having a higher dividend payout ratio. True or false?

The answer is false if clienteles exist. As long as enough high-dividend firms satisfy the dividend-loving investors, a firm won't be able to boost its share price by paying high dividends. An unsatisfied clientele must exist for this to happen, and there is no evidence that this is the case.

clientele effect

The observable fact that stocks attract particular groups based on dividend yield and the resulting tax effects.

Concept Questions

- 17.5a** How does the market react to unexpected dividend changes? What does this tell us about dividends? About dividend policy?
- 17.5b** What is a dividend clientele? All things considered, would you expect a risky firm with significant but highly uncertain growth prospects to have a low or a high dividend payout?

Stock Repurchases: An Alternative to Cash Dividends

17.6

Thus far in our chapter, we have considered cash dividends. Cash dividends are not the only way corporations distribute cash. Instead, a company can initiate a **stock repurchase**. Repurchases (or *buybacks*) have become an increasingly popular tool, and the amount spent on repurchases has become huge. For example, since 2010, S&P 500 companies have spent more than \$5.3 trillion on buybacks.

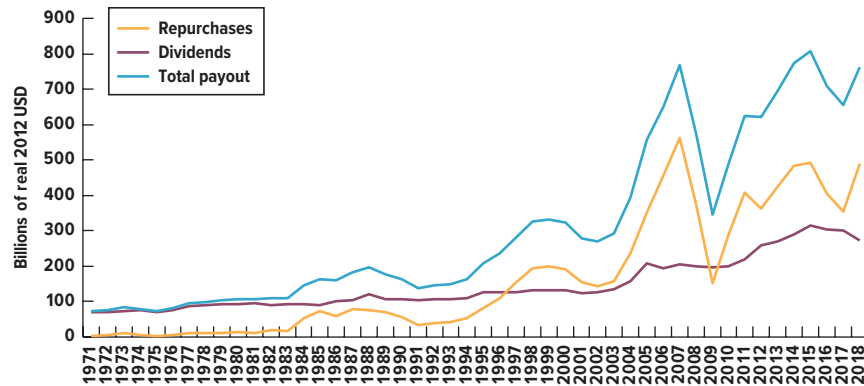
Another way to see how important repurchases have become is to compare them to cash dividends. Consider Figure 17.3, which shows the aggregate real (inflation-adjusted) dividends and stock repurchases by publicly held U.S. industrial firms for the period 1971–2018, along with the combined total. Aggregate real dividends have grown relatively

stock repurchase

The purchase, by a corporation, of its own shares of stock; also known as a *buyback*.

**FIGURE 17.3**

**Aggregate Real
(2012) Dividends and
Stock Repurchases
by Publicly Held U.S.
Industrial Firms:
1971–2018**



SOURCE: Redrawn by authors using Compustat data, following J. Farre-Mensa, R. Michaely, and M. Schmalz, "Payout Policy," *Annual Review of Financial Economics* 6, 2014, pp. 75–134.

steadily through time, but repurchases have exploded in the last two decades. They reached a peak in 2007 of about 2.5 times the size of aggregate dividends. Repurchases plunged in the 2008–2009 recession as firms conserved cash, but they rebounded in 2010.

Share repurchases are typically accomplished in one of three ways. First, companies may purchase their own stock, just as anyone would buy shares of a particular stock. In these *open market purchases*, the firm does not reveal itself as the buyer. The seller does not know whether the shares were sold back to the firm or sold to another investor.

Second, the firm could institute a *tender offer*. Here, the firm announces to all of its stockholders that it is willing to buy a fixed number of shares at a specific price. Suppose Arts and Crafts (A&C), Inc., has 1 million shares of stock outstanding, with a stock price of \$50 per share. The firm makes a tender offer to buy back 300,000 shares at \$60 per share. A&C chooses a price above \$50 to induce shareholders to sell—that is, tender—their shares. In fact, if the tender price is set high enough, shareholders may very well want to sell more than the 300,000 shares. In the extreme case where all outstanding shares are tendered, A&C will buy back 3 out of every 10 shares that a shareholder holds.

Finally, firms may repurchase shares from specific individual stockholders. This procedure has been called a *targeted repurchase*. For example, suppose the International Biotechnology Corporation purchased approximately 10 percent of the outstanding stock of the Prime Robotics Company (P-R Co.) in April at around \$38 per share. At that time, International Biotechnology announced to the Securities and Exchange Commission that it might eventually try to take control of P-R Co. In May, P-R Co. repurchased the International Biotechnology holdings at \$48 per share, well above the market price at that time. This offer was not extended to other shareholders.

CASH DIVIDENDS VERSUS REPURCHASE

Imagine an all-equity company with excess cash of \$300,000. The firm pays no dividends, and its net income for the year just ended is \$49,000. The market value balance sheet at the end of the year is represented here:

Market Value Balance Sheet (before paying out excess cash)			
Excess cash	\$ 300,000	Debt	\$ 0
Other assets	700,000	Equity	1,000,000
Total	<u>\$1,000,000</u>	Total	<u>\$1,000,000</u>

There are 100,000 shares outstanding. The total market value of the equity is \$1 million, so the stock sells for \$10 per share. Earnings per share (EPS) are $\$49,000/100,000 = \0.49 , and the price-earnings ratio (PE) is $\$10/\$0.49 = 20.4$.

One option the company is considering is a $\$300,000/100,000 = \3 per share extra cash dividend. Alternatively, the company is thinking of using the money to repurchase $\$300,000/\$10 = 30,000$ shares of stock.

If commissions, taxes, and other imperfections are ignored in our example, the stockholders shouldn't care which option is chosen. Does this seem surprising? It shouldn't, really. What is happening here is that the firm is paying out \$300,000 in cash. The new balance sheet is represented here:

Market Value Balance Sheet (after paying out excess cash)			
Excess cash	\$ 0	Debt	\$ 0
Other assets	<u>700,000</u>	Equity	<u>700,000</u>
Total	<u>\$700,000</u>	Total	<u>\$700,000</u>

If the cash is paid out as a dividend, there are still 100,000 shares outstanding, so each is worth \$7.

The fact that the per-share value fell from \$10 to \$7 is not a cause for concern. Consider a stockholder who owns 100 shares. At \$10 per share before the dividend, the total value is \$1,000.

After the \$3 dividend, this same stockholder has 100 shares worth \$7 each, for a total of \$700, plus $100 \times \$3 = \300 in cash, for a combined total of \$1,000. This illustrates what we saw early on: A cash dividend doesn't affect a stockholder's wealth if there are no imperfections. In this case, the stock price fell by \$3 when the stock went ex dividend.

Also, because total earnings and the number of shares outstanding haven't changed, EPS is still 49 cents. The price-earnings ratio falls to $\$7/\$0.49 = 14.3$. The reason we are looking at accounting earnings and PE ratios will be apparent in just a moment.

Alternatively, if the company repurchases 30,000 shares, there are 70,000 shares left outstanding. The balance sheet looks the same:

Market Value Balance Sheet (after share repurchase)			
Excess cash	\$ 0	Debt	\$ 0
Other assets	<u>700,000</u>	Equity	<u>700,000</u>
Total	<u>\$700,000</u>	Total	<u>\$700,000</u>

The company is worth \$700,000 again, so each remaining share is worth $\$700,000/70,000 = \10 . Our stockholder with 100 shares is obviously unaffected. For example, if she was so inclined, she could sell 30 shares and end up with \$300 in cash and \$700 in stock, just as she has if the firm pays the cash dividend. This is another example of a homemade dividend.

In this second case, EPS goes up because total earnings remain the same while the number of shares goes down. The new EPS is $\$49,000/70,000 = \0.70 . The important thing to notice is that the PE ratio is $\$10/\$0.70 = 14.3$, as it was following the dividend.

This example illustrates the important point that, if there are no imperfections, a cash dividend and a share repurchase are essentially the same thing. This is another illustration of dividend policy irrelevance when there are no taxes or other imperfections.

REAL-WORLD CONSIDERATIONS IN A REPURCHASE

The example we have just described shows that a repurchase and a cash dividend are the same thing in a world without taxes and transaction costs. In the real world, there are some accounting differences between a share repurchase and a cash dividend, but the most important difference is in the tax treatment.

Under current tax law, a repurchase has a significant tax advantage over a cash dividend. A dividend is taxed, and a shareholder has no choice about whether or not to receive the dividend. In a repurchase, a shareholder pays taxes only if (1) the shareholder actually chooses to sell and (2) the shareholder has a capital gain on the sale.

Suppose a dividend of \$1 per share is taxed at ordinary rates. Investors in the 28 percent tax bracket who own 100 shares of the security pay $\$100 \times .28 = \28 in taxes. Selling shareholders would pay far lower taxes if \$100 worth of stock were repurchased. This is because taxes are paid only on the profit from a sale. The gain on a sale would be only \$40 if shares sold at \$100 were originally purchased at \$60. The capital gains tax would be $.28 \times \$40 = \11.20 . Note that the 2003 reductions in dividend and capital gains tax rates do not change the fact that a repurchase has a potentially large tax edge.

To give a few recent examples, we earlier mentioned the Microsoft \$40 billion buyback announced in 2019, but the company also repurchased about 150 million shares of its stock during 2019, totaling about \$20 billion. Apple completed \$67 billion in buybacks in 2019 and then announced another \$40 billion buyback. About the same time, Bank of America announced that it would buy back \$30.9 billion of its stock over the next year. Of course, buybacks are not limited to the United States. In January 2020, German sporting goods company Adidas announced a €1 billion buyback.

IBM is well known for its aggressive repurchasing policies. Since 2003, the company had spent about \$147.5 billion on buybacks through the end of 2019. In fact, from the beginning of 2003 through the end of 2019, the number of shares in the company fell from 1.72 billion shares to 886 million shares. In late 2019, the company announced that it would suspend further buybacks in order to pay for its \$34 billion acquisition of Red Hat.

One thing to note is that not all announced stock repurchase plans are completed. Based on research covering the 2004–2007 period, average completion rates were only 81 percent.⁴

SHARE REPURCHASE AND EPS

You may read in the popular financial press that a share repurchase is beneficial because it causes earnings per share to increase. As we have seen, this will happen. The reason is that a share repurchase reduces the number of outstanding shares, but it has no effect on total earnings. As a result, EPS rises.

⁴See A. A. Bonaimé, “Mandatory Disclosure and Firm Behavior: Evidence from Share Repurchases,” *The Accounting Review* 90 (2015).

The financial press may place undue emphasis on EPS figures in a repurchase agreement. In our preceding example, we saw that the value of the stock wasn't affected by the EPS change. In fact, the PE ratio was exactly the same when we compared a cash dividend to a repurchase.

Concept Questions

- 17.6a** Why might a stock repurchase make more sense than an extra cash dividend?
- 17.6b** What is the effect of a stock repurchase on a firm's EPS? Its PE?

What We Know and Do Not Know about Dividend and Payout Policies

17.7

DIVIDENDS AND DIVIDEND PAYERS

As we have discussed, there are numerous good reasons favoring a dividend policy of low (or no) payout. Nonetheless, in the United States, aggregate dividends paid are quite large. For example, in 1978, U.S. industrial firms listed on the major exchanges paid \$31.3 billion in total dividends. By 2000, that number had risen to \$101.6 billion (unadjusted for inflation), an increase of over 200 percent (after adjusting for inflation, the increase is smaller, 22.7 percent, but still substantial).

While we know dividends are large in the aggregate, we also know that the number of companies that pay dividends has declined. Over the same 1978–2000 period, the number of industrial companies paying dividends declined from over 2,000 to just under 1,000, and the percentage of these firms paying dividends declined 65 percent, to 19 percent.⁵

The fact that aggregate dividends grew while the number of payers fell so sharply seems a bit paradoxical, but the explanation is straightforward: Dividend payments are heavily concentrated in a relatively small set of large firms. In 2000, for example, about 80 percent of aggregate dividends were paid by just 100 firms. The top 25 payers, which include such well-known giants as ExxonMobil and General Electric, collectively paid about 55 percent of all dividends. The reason that dividends grew while dividend payers shrank is that the decline in dividend payers is almost entirely due to smaller firms, which tend to pay smaller dividends in the first place.

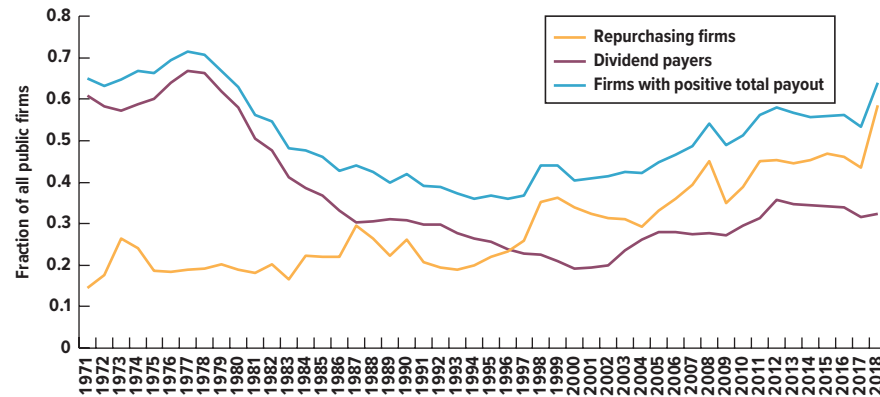
One important reason that the percentage of dividend-paying firms has declined is that the population of firms has changed. There has been a huge increase in the number of newly listed firms over the last 25 or so years. Newly listed firms tend to be younger and less profitable. Such firms need their internally generated cash to fund growth and typically do not pay dividends.

Another factor at work is that firms appear to be more likely to begin making payouts using share repurchases, which are flexible, rather than committing to making cash distributions. Such a policy seems quite sensible given our previous discussions. However, after

⁵These figures and those in the following paragraph are from H. DeAngelo, L. DeAngelo, and D. J. Skinner, "Are Dividends Disappearing? Dividend Concentration and the Consolidation of Earnings," *Journal of Financial Economics* 72 (2004).

**FIGURE 17.4**

Proportion of Dividend Payers, Repurchasers, and Firms with Positive Total Payout among All Publicly Held U.S. Industrial Firms: 1971–2018



SOURCE: Redrawn by authors using Compustat data, following J. Farre-Mensa, R. Michaely, and M. Schmalz, "Payout Policy," *Annual Review of Financial Economics* 6 (2014), pp. 75–134.

controlling for the changing mix of firms and the increase in share repurchasing activity, there still appears to be a decreased propensity to pay dividends among certain types of older, better established firms, though further research is needed on this question.

The fact that the number of dividend-paying firms has declined so sharply is an interesting phenomenon. Making matters even more interesting is evidence showing that the trend may have begun to reverse itself. Take a look at Figure 17.4, which shows the percentage of industrial firms paying dividends over the period 1971–2018, along with the percentage of (1) firms doing repurchases and (2) firms with a positive payout of one type or the other (or both). As shown, there is a pronounced downward trend, but that trend appears to bottom out in 2000 and then sharply reverse in 2002. So what's going on?

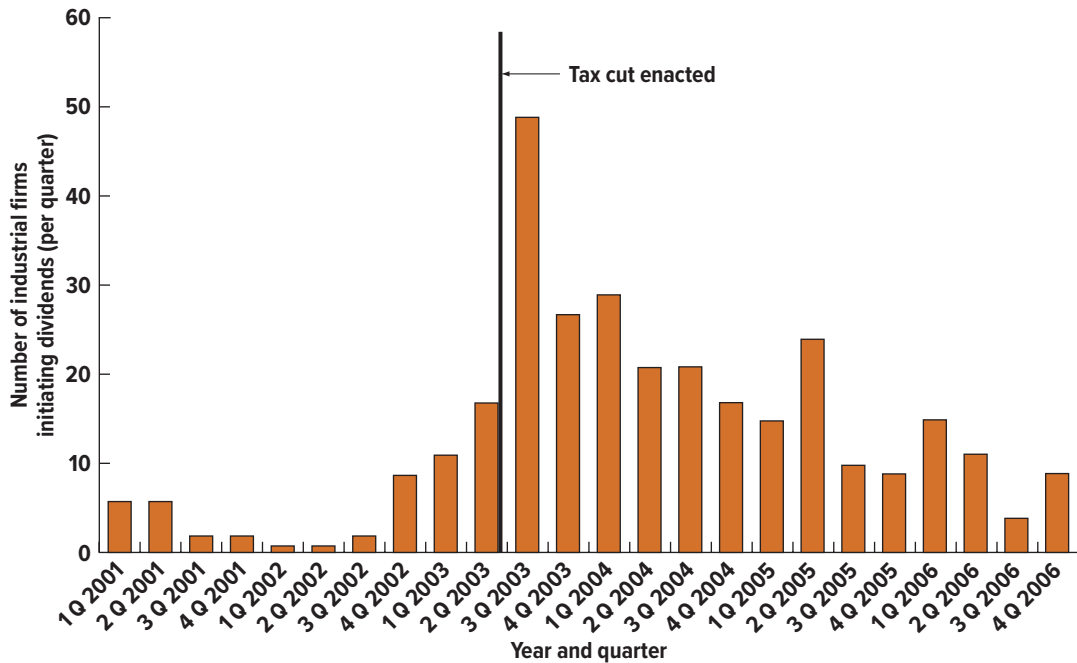
Part of the apparent rebound in Figure 17.4 is probably an illusion. The percentage of firms paying dividends rose because nonpayers dropped out in large numbers.⁶ The number of firms listed on the major stock markets dropped sharply, from over 5,000 to under 4,000, during the period 2000–2005. About 2,000 firms delisted over this period, 98 percent of which were not dividend payers. By 2013, the number of listed firms had declined to below 3,000, and the percentage of dividend payers reached 36 percent.

Once we control for the drop-out problem, there is still an increase in the number of dividend payers, but it happens in 2003. As shown in Figure 17.5, the uptick is concentrated in the months following May 2003. What is so special about this month? The answer is that in May 2003, top personal tax rates on dividends were slashed from about 38 to 15 percent. Consistent with our earlier tax arguments, a reduction in personal tax rates led to increases in dividends.

It is important not to read too much into Figure 17.5. It seems clear that the reduction in tax rates did have an effect, but, on balance, what we see is a few hundred firms initiating dividends. There are still thousands of firms that did not initiate dividends, even though the tax rate reduction was very large. The evidence suggests that tax rates matter, but they are not a primary determinant of dividend policy. This interpretation is consistent with the results of a 2005 survey of financial executives, more than two-thirds of whom said that the tax rate cut probably or definitely would not affect their dividend policies.⁷

⁶These numbers and this explanation are from R. Chetty and E. Saez, "The Effects of the 2003 Dividend Tax Cut on Corporate Behavior: Interpreting the Evidence," *American Economic Review, Papers and Proceedings* 96 (2006).

⁷See A. P. Brav, J. R. Graham, C. R. Harvey, and R. Michaely, "Managerial Response to the May 2003 Dividend Tax Cut," *Financial Management* 37 (2008), pp. 611–24.

**FIGURE 17.5** Regular Dividend Initiations, 2001–2006

SOURCE: A. P. Brav, J. R. Graham, C. R. Harvey, and R. Michaely, "Managerial Response to the May 2003 Dividend Tax Cut," *Financial Management* 37 (2008), pp. 611–24.

A second force that may be at work over time is the maturing of many of the (surviving) newly listed firms we mentioned earlier. As these firms have become better established, their profitability has increased (and, potentially, their investment opportunities have decreased), and they have begun to pay dividends.

A third factor that may be contributing to the increase in the number of dividend payers is a little more subtle. The technology-heavy Nasdaq index plummeted in the spring of 2000 (due to the “dot-com” crash), and it became clear that many newly listed companies were likely to fail. Shortly thereafter, major accounting scandals at companies such as Enron and WorldCom left investors unsure of the trustworthiness of reported earnings. In such an environment, companies may have chosen to initiate dividends in an attempt to signal to investors that they had the cash to make dividend payments now and in the future.

The apparent reversal in the decline of dividend payers is a recent phenomenon, so its significance remains to be seen. It may prove to be just a transient event in the middle of a long decline. We will have to wait and see.

CORPORATIONS SMOOTH DIVIDENDS

As we previously observed, dividend cuts are frequently viewed as very bad news by market participants. As a result, companies only cut dividends when there is no other acceptable alternative. For the same reason, companies are also reluctant to increase dividends unless they are sure the new dividend level can be sustained.

In practice, what we observe is that dividend-paying companies tend to raise dividends only after earnings have risen, and they don’t increase or cut dividends in response to

temporary earnings fluctuations. In other words, (1) dividend growth lags earnings growth, and (2) dividend growth will tend to be much smoother than earnings growth.

To see how important dividend stability and steady growth are to financial managers, consider that, in 2019, 1,808 companies in the United States increased their dividend payments, while only 309 companies decreased their dividend payments. Two companies with long histories of dividend increases are Procter & Gamble and Colgate-Palmolive. At the end of 2019, Procter & Gamble had increased its dividend for 64 consecutive years, and Colgate-Palmolive had increased its dividend for 57 consecutive years. Overall, as of 2019, 56 companies in the S&P 500 had increased dividends for at least 25 consecutive years.

PUTTING IT ALL TOGETHER

Much of what we have discussed in this chapter (and much of what we know about dividends from decades of research) can be pulled together and summarized in the following five observations:⁸

1. Aggregate dividend and stock repurchases are massive, and they have increased steadily in nominal and real terms over the years.
2. Dividends are heavily concentrated among a relatively small number of large, mature firms.
3. Managers are very reluctant to cut dividends, normally doing so only due to firm-specific problems.
4. Managers smooth dividends, raising them slowly and incrementally as earnings grow.
5. Stock prices react to unanticipated changes in dividends.

The challenge now is to fit these five pieces into a reasonably coherent picture. With regard to payouts in general, meaning the combination of stock repurchases and cash dividends, a simple life cycle theory fits Points 1 and 2. The key ideas are straightforward. First, relatively young and less profitable firms generally should not make cash distributions. They need the cash to fund investments (and flotation costs discourage the raising of outside cash).

As a firm matures, it begins to generate free cash flow (which, you will recall, is internally generated cash flow beyond that needed to fund profitable investment activities). Significant free cash flow can lead to agency problems if it is not distributed. Managers may become tempted to pursue empire building or otherwise spend the excess cash in ways not in the shareholders' best interests. Firms come under pressure to make distributions rather than hoard cash. And, consistent with what we observe, we expect large firms with a history of profitability to make large distributions.

The life-cycle theory says that firms trade off the agency costs of excess cash retention against the potential future costs of external equity financing. A firm should begin making distributions when it generates sufficient internal cash flow to fund its investment needs now and into the foreseeable future.

The more complex issue concerns the type of distribution, cash dividends versus repurchases. The tax argument in favor of repurchases is a clear and strong one. Further, repurchases

⁸This list is distilled in part from a longer list in H. DeAngelo and L. DeAngelo, "Payout Policy Pedagogy: What Matters and Why," *European Financial Management* 13 (2007).

are a much more flexible option (and managers greatly value financial flexibility), so the question is: Why would firms ever choose a cash dividend?

If we are to answer this question, we have to ask a different question. What can a cash dividend accomplish that a share repurchase cannot accomplish? One answer is that when a firm makes a commitment to pay a cash dividend now and into the future, it sends a two-part signal to the markets. As we have already discussed, one signal is that the firm anticipates being profitable, with the ability to make the payments on an ongoing basis. Note that a firm cannot benefit by trying to fool the market in this regard because the firm would ultimately be punished when it couldn't make the dividend payment (or couldn't make it without relying on external financing). A cash dividend may let a firm distinguish itself from less profitable rivals.

A second, and more subtle, signal takes us back to the agency problem of free cash flow. By committing to pay cash dividends now and in the future, the firm signals that it won't be hoarding cash (or at least not as much cash), thereby reducing agency costs and enhancing shareholder wealth.

This two-part signaling story is consistent with Points 3–5 above, but an obvious objection remains: Why don't firms just commit to a policy of setting aside whatever money would be used to pay dividends and use it instead to buy back shares? After all, either way, a firm is committing to pay out cash to shareholders.

A fixed repurchase strategy suffers from two drawbacks. The first is verifiability. A firm could announce an open market repurchase and then not do it. By suitably fudging its books, it would be some time before the deception was discovered. It would be necessary for shareholders to develop a monitoring mechanism, meaning some sort of way for stockholders to know for sure that the repurchase was in fact done. Such a mechanism wouldn't be difficult to build (it could be a simple trustee relationship such as we observe in the bond markets), but it currently does not exist. Of course, a tender offer repurchase needs little or no verification, but such offers have expenses associated with them. The beauty of a cash dividend is that it needs no monitoring. A firm is forced to cut and mail checks four times per year, year in and year out.

A second objection to a fixed repurchase strategy is more controversial. Suppose managers, as insiders, are better able than stockholders to judge whether their stock price is too high or too low. (Note that this idea does not conflict with semistrong market efficiency if inside information is the reason.) In this case, a fixed repurchase commitment forces management to buy back stock even in circumstances when the stock is overvalued. In other words, it forces management into making negative NPV investments.

More research on the cash dividend versus share repurchase question is needed, but the historical trend seems to be favoring continued growth in repurchases relative to dividends.

Total corporate payouts seem to be relatively stable over time at roughly 20 percent of aggregate earnings (see Figure 17.3), but repurchases are becoming a larger portion of that total. The split reached about 50–50 in the latter part of the 1990s, but it looks like aggregate repurchases have recently passed aggregate dividends.

One aspect of aggregate cash dividends that has not received much attention is that there may be a strong legacy effect. Before 1982, the regulatory status of stock repurchases was somewhat murky, creating a significant disincentive. In 1982, the SEC, after years of debate, created a clear set of guidelines for firms to follow, thereby making repurchases much more attractive.

The legacy effect arises because many of the giant firms that pay such a large portion of aggregate dividends were paying dividends before (and perhaps long before) 1982. To the extent that these firms are unwilling to cut their dividends, aggregate cash dividends

will be large, but only because of a “lock-in” effect for older firms. If locked-in, legacy payers account for much of the aggregate dividend, what we should observe is (1) a sharply reduced tendency for maturing firms to initiate dividends and (2) a growth in repurchases relative to cash dividends over time. We actually do see evidence of both of these trends; legacy effects alone can’t account for all cash dividend payers.

The Pros and Cons of Paying Dividends	
Pros	Cons
1. Cash dividends can underscore good results and provide support to the stock price.	1. Dividends are taxed to recipients.
2. Dividends may attract institutional investors who prefer some return in the form of dividends. A mix of institutional and individual investors may allow a firm to raise capital at a lower cost because of the ability of the firm to reach a wider market.	2. Dividends can reduce internal sources of financing. Dividends may force the firm to forgo positive NPV projects or to rely on costly external equity financing.
3. Stock price usually increases with the announcement of a new or increased dividend.	3. Once established, dividend cuts are hard to make without adversely affecting a firm’s stock price.
4. Dividends absorb excess cash flow and may reduce agency costs that arise from conflicts between management and shareholders.	

SOME SURVEY EVIDENCE ON DIVIDENDS

A study surveyed a large number of financial executives regarding dividend policy. One of the questions asked was, “Do these statements describe factors that affect your company’s dividend decisions?” Table 17.1 shows some of the results.

As shown in Table 17.1, financial managers are very disinclined to cut dividends. Moreover, they are very conscious of their previous dividends and desire to maintain a relatively steady dividend. In contrast, the cost of external capital and the desire to attract “prudent man” investors (those with fiduciary duties) are less important.

 **TABLE 17.1**
Survey Responses on
Dividend Decisions*

Do These Statements Affect Your Dividend Policy?	Percent Who Agree or Strongly Agree
1. Avoiding dividend cuts	93.8%
2. Maintaining a “smooth” dividend over time	89.6
3. Size of recently paid dividends	88.2
4. Making sure that any changes we make won’t have to be reversed later	77.9
5. The rate at which dividends are changing or growing	66.7
6. The cost of cutting dividends exceeds the cost of raising external capital	42.8
7. Using dividends to attract investors subject to “prudent man” investment restrictions	41.7

*Survey respondents were asked the question, “Do these statements describe factors that affect your company’s dividend decisions?”
SOURCE: Adapted from Table 4 of A. Brav, et al., “Payout Policy in the 21st Century,” *Journal of Financial Economics*, 2005.



IN THEIR OWN WORDS . . .

Fischer Black on Why Firms Pay Dividends

I think investors like dividends. They believe that dividends enhance stock value (given the firm's prospects), and they feel uncomfortable spending out of their capital.

We see evidence for this everywhere: Investment advisers and institutions treat a high-yield stock as both attractive and safe, financial analysts value a stock by predicting and discounting its dividends, financial economists study the relation between stock prices and actual dividends, and investors complain about dividend cuts.

What if investors were neutral toward dividends? Investment advisers would tell clients to spend indifferently from income and capital and, if taxable, to avoid income; financial analysts would ignore dividends in valuing stocks; financial economists would treat stock price and the discounted value of dividends as equal, even when stocks are mispriced; and a firm would apologize to its taxable investors when forced by an accumulated earnings tax to pay dividends. This is not what we observe.

Furthermore, changing dividends seems a poor way to tell the financial markets about a firm's prospects. Public statements can better detail the firm's prospects and have more impact on both the speaker's and the firm's reputations.

I predict that under current tax rules, dividends will gradually disappear.

The late Fischer Black was a partner at Goldman, Sachs & Co., an investment banking firm. Before that, he was a professor of finance at MIT. He is one of the fathers of option pricing theory, and he is widely regarded as one of the preeminent financial scholars. He is well known for his creative ideas, many of which were dismissed at first only to become part of accepted lore when others finally came to understand them. He is sadly missed by his colleagues.

Table 17.2 is drawn from the same survey, but here the responses are to the question, "How important are the following factors to your company's dividend decision?" Not surprisingly given the responses in Table 17.1 and our earlier discussion, the highest priority is maintaining a consistent dividend policy. The next several items are also consistent with our previous analysis. Financial managers are very concerned about earnings stability and future earnings levels in making dividend decisions, and they consider the availability of good investment opportunities. Survey respondents also believed that attracting both institutional and individual (retail) investors was relatively important.

In contrast to our discussion of taxes and flotation costs in the earlier part of this chapter, the financial managers in this survey did not think that personal taxes paid on dividends by shareholders were very important. Even fewer financial managers thought that equity flotation costs were relevant.

How Important Are the Following?	Percent Who Think This Is Important or Very Important
1. Being consistent with our historic dividend policy	84.1%
2. Future earnings stability	71.9
3. Sustainable earnings changes	67.1
4. Attracting institutional investors	52.5
5. Good quality investments	47.6
6. Attracting retail investors	44.5
7. Personal taxes paid by our shareholders on dividends we pay	21.1
8. Costs associated with issuing new equity	9.3

*Survey respondents were asked the question, "How important are the following factors to your company's dividend decision?"

SOURCE: Adapted from Table 5 of A. Brav, et al., "Payout Policy in the 21st Century," *Journal of Financial Economics*, 2005.



TABLE 17.2
Survey Responses on
Dividend Decisions*

17.8 Stock Dividends and Stock Splits

stock dividend

A payment made by a firm to its owners in the form of stock, diluting the value of each share outstanding.

stock split

An increase in a firm's shares outstanding without any change in owners' equity.

Another type of dividend is paid out in shares of stock. This type of dividend is called a **stock dividend**. A stock dividend is not a true dividend because it is not paid in cash. The effect of a stock dividend is to increase the number of shares that each owner holds. Because there are more shares outstanding, each is worth less.

A stock dividend is commonly expressed as a percentage; for example, a 20 percent stock dividend means that a shareholder receives one new share for every five currently owned (a 20 percent increase). Because every shareholder receives 20 percent more stock, the total number of shares outstanding rises by 20 percent. As we will see in a moment, the result is that each share of stock is worth about 20 percent less.

A **stock split** is essentially the same thing as a stock dividend, except that a split is expressed as a ratio instead of a percentage. When a split is declared, each share is split up to create additional shares. For example, in a 3-for-1 stock split, each old share is split into three new shares.

SOME DETAILS ABOUT STOCK SPLITS AND STOCK DIVIDENDS

Stock splits and stock dividends have essentially the same impacts on the corporation and the shareholder: They increase the number of shares outstanding and reduce the value per share. The accounting treatment is not the same, and it depends on two things: (1) whether the distribution is a stock split or a stock dividend and (2) the size of the stock dividend if it is called a dividend. In recent years, stock splits have slowed considerably. During the 1990s, an average of 64 S&P 500 companies split each year. From 2008 through 2012, the number fell to about 12 per year. In 2018, only five S&P 500 companies split their shares.

By convention, stock dividends of less than 20 to 25 percent are called *small stock dividends*. The accounting procedure for such a dividend is discussed next. A stock dividend greater than this range of 20 to 25 percent is called a *large stock dividend*. Large stock dividends are not uncommon. For example, in April 2019, First Financial Bankshares announced a 100 percent stock dividend in the form of a 2-for-1 stock split. Except for some relatively minor accounting differences, a stock dividend has the same effect as a stock split.



Information on upcoming stock splits is available on the splits calendar at www.investmenthouse.com and finance.yahoo.com.

Example of a Small Stock Dividend The Peterson Co., a consulting firm specializing in difficult accounting problems, has 10,000 shares of stock outstanding, each selling at \$66. The total market value of the equity is $\$66 \times 10,000 = \$660,000$. With a 10 percent stock dividend, each stockholder receives one additional share for each 10 owned, and the total number of shares outstanding after the dividend is 11,000.

Before the stock dividend, the equity portion of Peterson's balance sheet might look like this:

Common stock (\$1 par, 10,000 shares outstanding)	\$ 10,000
Capital in excess of par value	200,000
Retained earnings	290,000
Total owners' equity	<u>\$500,000</u>

A seemingly arbitrary accounting procedure is used to adjust the balance sheet after a small stock dividend. Because 1,000 new shares are issued, the common stock account is increased by \$1,000 (1,000 shares at \$1 par value each), for a total of **\$11,000**. The market

price of \$66 is \$65 greater than the par value, so the “excess” of $\$65 \times 1,000$ shares = \$65,000 is added to the capital surplus account (capital in excess of par value), producing a total of **\$265,000**.

Total owners’ equity is unaffected by the stock dividend because no cash has come in or out, so retained earnings are reduced by the entire \$66,000, leaving **\$224,000**. The net effect of these machinations is that Peterson’s equity accounts now look like this:

Common stock (\$1 par, 11,000 shares outstanding)	\$ 11,000
Capital in excess of par value	265,000
Retained earnings	224,000
Total owners’ equity	\$500,000

Example of a Stock Split A stock split is conceptually similar to a stock dividend, but it is commonly expressed as a ratio. For example, in a 3-for-2 split, each shareholder receives one additional share of stock for each two held originally, so a 3-for-2 split amounts to a 50 percent stock dividend. Again, no cash is paid out, and the percentage of the entire firm that each shareholder owns is unaffected.

The accounting treatment of a stock split is a little different from (and simpler than) that of a stock dividend. Suppose Peterson decides to declare a 2-for-1 stock split. The number of shares outstanding will double to **20,000**, and the par value will be halved to **\$.50** per share. The owners’ equity after the split is represented as follows:

Common stock (\$.50 par, 20,000 shares outstanding)	\$ 10,000
Capital in excess of par value	200,000
Retained earnings	290,000
Total owners’ equity	\$500,000



For a list of recent stock splits, try www.nasdaq.com/market-activity/stock-splits.

Note that, for all three of the categories, the figures on the right are completely unaffected by the split. The only changes are in the par value per share and the number of shares outstanding. Because the number of shares has doubled, the par value of each is cut in half.

Example of a Large Stock Dividend In our example, if a 100 percent stock dividend were declared, 10,000 new shares would be distributed, so **20,000** shares would be outstanding. At a **\$1** par value per share, the common stock account would rise by \$10,000, for a total of **\$20,000**. The retained earnings account would be reduced by \$10,000, leaving **\$280,000**. The result would be the following:

Common stock (\$1 par, 20,000 shares outstanding)	\$ 20,000
Capital in excess of par value	200,000
Retained earnings	280,000
Total owners’ equity	\$500,000

VALUE OF STOCK SPLITS AND STOCK DIVIDENDS

The laws of logic tell us that stock splits and stock dividends can (1) leave the value of the firm unaffected, (2) increase its value, or (3) decrease its value. Unfortunately, the issues are complex enough that we cannot easily determine which of the three relationships holds.

The Benchmark Case A strong case can be made that stock dividends and splits do not change either the wealth of any shareholder or the wealth of the firm as a whole. In our

preceding example, the equity had a total market value of \$660,000. With the small stock dividend, the number of shares increased to 11,000, so it seems that each would be worth $\$660,000/11,000 = \60 .

For example, a shareholder who had 100 shares worth \$66 each before the dividend would have 110 shares worth \$60 each afterward. The total value of the stock is \$6,600 either way; so the stock dividend doesn't really have any economic effect.

After the stock split, there are 20,000 shares outstanding, so each should be worth $\$660,000/20,000 = \33 . In other words, the number of shares doubles and the price halves. From these calculations, it appears that stock dividends and splits are just paper transactions.

Although these results are relatively obvious, reasons are often given to suggest that there may be some benefits to these actions. The typical financial manager is aware of many real-world complexities; for that reason, the stock split or stock dividend decision is not treated lightly in practice.

Popular Trading Range Proponents of stock dividends and stock splits frequently argue that a security has a proper **trading range**. When the security is priced above this level, many investors do not have the funds to buy the common trading unit of 100 shares, called a *round lot*. Although securities can be purchased in *odd-lot* form (fewer than 100 shares), the commissions are greater. Firms will split the stock to keep the price in this trading range.

For example, Microsoft has split nine times since the company went public in 1986. The stock has split 3-for-2 on two occasions and 2-for-1 a total of seven times. So for every share of Microsoft you owned in 1986 when the company first went public, you would own 288 shares as of the most recent stock split. Similarly, since Walmart went public in 1970, it has split its stock 2-for-1 eleven times, and Apple has split 7-for-1 once and 2-for-1 three times since going public in 1980. For a really long split history, consider Procter & Gamble, which has split 5-for-1 twice, 1.5-to-1 once, and 2-for-1 eight times since 1920. Each share of P&G purchased prior to the company's first stock split would now be worth 9,600 shares.

Although this argument is a popular one, its validity is questionable for a number of reasons. Mutual funds, pension funds, and other institutions have steadily increased their trading activity since World War II and now handle a sizable percentage of total trading volume (on the order of 90 percent of NYSE trading volume, for example). Because these institutions buy and sell in huge amounts, the individual share price is of little concern.

Furthermore, we sometimes observe share prices that are quite large that do not appear to cause problems. To take a well-known case, Berkshire Hathaway, a widely respected company headed by legendary investor Warren Buffett, sold for about \$345,000 per share in early 2020.

Finally, there is evidence that stock splits may actually decrease the liquidity of the company's shares. Following a 2-for-1 split, the number of shares traded should more than double if liquidity is increased by the split. This doesn't appear to happen, and the reverse is sometimes observed.

REVERSE SPLITS

A less frequently encountered financial maneuver is the **reverse split**. For example, in December 2019, Ascena Retail Group, parent of Ann Taylor and Loft, announced a 1-for-20 reverse stock split. Also in December 2019, Recon Technology underwent a 1-for-5 reverse split. In a 1-for-5 reverse split, each investor exchanges five old shares for one new share. The par value is increased by a factor of 5 in the process. In what was probably one of the biggest reverse splits ever (in terms of market cap), banking giant Citigroup

trading range

The price range between the highest and lowest prices at which a stock is traded.

reverse split

A stock split in which a firm's number of shares outstanding is reduced.

announced in March 2011 that it would do a 1-for-10 reverse split, thereby reducing the number of shares outstanding from 29 billion to 2.9 billion. As with stock splits and stock dividends, a case can be made that a reverse split has no real effect.

Given real-world imperfections, three related reasons are cited for reverse splits. First, transaction costs to shareholders may be less after the reverse split. Second, the liquidity and marketability of a company's stock might be improved when its price is raised to the popular trading range. Third, stocks selling at prices below a certain level are not considered respectable, meaning that investors underestimate these firms' earnings, cash flow, growth, and stability. Some financial analysts argue that a reverse split can achieve instant respectability. As was the case with stock splits, none of these reasons is particularly compelling, especially not the third one.

There are two other reasons for reverse splits. First, stock exchanges have minimum price per share requirements. A reverse split may bring the stock price up to such a minimum. For example, Nasdaq begins the delisting process for companies whose stock price drops below \$1 per share for 30 days. Following the collapse of the internet boom in 2001–2002, a large number of internet-related companies found themselves in danger of being delisted and used reverse splits to boost their stock prices. In 2019, both Rite Aid and Blue Apron executed reverse splits in order to remain listed on the NYSE. Second, companies sometimes perform reverse splits and, at the same time, buy out any stockholders who end up with less than a certain number of shares.

For example, in October 2019, FitLife Brands completed a reverse/forward split. In this case, the company first did a 1-for-8,000 reverse stock split and then repurchased all shares held by stockholders with less than one share of stock, eliminating smaller shareholders thus reducing the total number of shareholders. What made the proposal especially imaginative was that immediately after the reverse split, the company did an 800-for-1 ordinary split. The net effect for remaining shareholders was the same as a 1-for-10 reverse stock split.

Concept Questions

- 17.8a** What is the effect of a stock split on stockholder wealth?
- 17.8b** How does the accounting treatment of a stock split differ from that used with a small stock dividend?

Summary and Conclusions

17.9

In this chapter, we first discussed the types of dividends and how they are paid. We then defined dividend policy and examined whether or not dividend policy matters. Next, we illustrated how a firm might establish a dividend policy and described an important alternative to cash dividends, a share repurchase.

In covering these subjects, we discussed the following points:

1. Dividend policy is irrelevant when there are no taxes or other imperfections because shareholders can effectively undo the firm's dividend strategy. Shareholders who receive dividends greater than desired can reinvest the excess. Conversely, shareholders who receive dividends smaller than desired can sell off extra shares of stock.

2. Individual shareholder income taxes and new issue flotation costs are real-world considerations that favor a low dividend payout. With taxes and new issue costs, the firm should pay out dividends only after all positive NPV projects have been fully financed.
3. There are groups in the economy that may favor a high payout. These include many large institutions such as pension plans. Recognizing that some groups prefer a high payout and some prefer a low payout, the clientele effect argument supports the idea that dividend policy responds to the needs of stockholders. For example, if 40 percent of the stockholders prefer low dividends and 60 percent of the stockholders prefer high dividends, approximately 40 percent of companies will have a low dividend payout, and 60 percent will have a high payout. This sharply reduces the impact of any individual firm's dividend policy on its market price.
4. A stock repurchase acts much like a cash dividend, but it has a significant tax advantage. Stock repurchases are therefore a very useful part of overall dividend policy.
5. We discussed recent research and thinking on dividend policy. We saw that dividends are heavily concentrated in a relatively small number of larger, older firms and that the use of share repurchases continues to grow. We described a simple life cycle theory of distributions in which firms trade off the agency costs of excess cash retention against the future costs of external equity financing. The implication is that younger firms with significant growth opportunities will not distribute cash, but older, profitable firms with significant free cash flow will distribute cash.

To close our discussion of dividends, we emphasize one last time the difference between dividends and dividend policy. Dividends are important because the value of a share of stock is ultimately determined by the dividends that will be paid. What is less clear is whether the time pattern of dividends (more now versus more later) matters. This is the dividend policy question, and it is not easy to give a definitive answer to it.

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Can you answer the following *Connect Quiz* questions?

Section 17.1 Dividends are paid to the parties listed as shareholders on what date?

Section 17.3 What factors favor a high dividend payout?

Section 17.4 Which parties are most apt to prefer a low dividend payout?

Section 17.8 Tomas currently owns 300 shares of Doo Little Corp. How many shares will he own if the firm does a 4-for-5 reverse stock split?

CONCEPTS REVIEW AND CRITICAL THINKING QUESTIONS

1. **Dividend Policy Irrelevance [LO2]** How is it possible that dividends are so important, but at the same time, dividend policy is irrelevant?
2. **Stock Repurchases [LO4]** What is the impact of a stock repurchase on a company's debt ratio? Does this suggest another use for excess cash?
3. **Dividend Chronology [LO1]** On Tuesday, December 5, Hometown Power Co.'s board of directors declares a dividend of 75 cents per share payable on Wednesday,

January 17, to shareholders of record as of Wednesday, January 3. When is the ex-dividend date? If a shareholder buys stock before that date, who gets the dividends on those shares, the buyer or the seller?

4. **Alternative Dividends [LO1]** Some corporations, like one British company that offers its large shareholders free crematorium use, pay dividends in kind (that is, offer their services to shareholders at below-market cost). Should mutual funds invest in stocks that pay these dividends in kind? (The fundholders do not receive these services.)
5. **Dividends and Stock Price [LO1]** If increases in dividends tend to be followed by (immediate) increases in share prices, how can it be said that dividend policy is irrelevant?
6. **Dividends and Stock Price [LO1]** Last month, Central Virginia Power Company, which had been having trouble with cost overruns on a nuclear power plant that it had been building, announced that it was “temporarily suspending payments due to the cash flow crunch associated with its investment program.” The company’s stock price dropped from \$28.50 to \$25 when this announcement was made. How would you interpret this change in the stock price (that is, what would you say caused it)?
7. **Dividend Reinvestment Plans [LO1]** The DRK Corporation has recently developed a dividend reinvestment plan, or DRIP. The plan allows investors to reinvest cash dividends automatically in DRK in exchange for new shares of stock. Over time, investors in DRK will be able to build their holdings by reinvesting dividends to purchase additional shares of the company.

A large number of companies offer dividend reinvestment plans. Most companies with DRIPs charge no brokerage or service fees. In fact, the shares of DRK will be purchased at a 10 percent discount from the market price.

A consultant for DRK estimates that about 75 percent of DRK’s shareholders will take part in this plan. This is somewhat higher than the average.

Evaluate DRK’s dividend reinvestment plan. Will it increase shareholder wealth? Discuss the advantages and disadvantages involved here.

8. **Dividend Policy [LO2]** For initial public offerings of common stock, 2017 was a slow year, with about \$24.53 billion raised by the process. Relatively few of the 108 firms involved paid cash dividends. Why do you think that most chose not to pay cash dividends?

Use the following information to answer the next two questions:

Historically, the U.S. tax code treated dividend payments made to shareholders as ordinary income. Thus, dividends were taxed at the investor’s marginal tax rate, which was as high as 38.6 percent in 2002. Capital gains were taxed at a capital gains tax rate, which was the same for most investors and fluctuated through the years. In 2002, the capital gains tax rate stood at 20 percent. In an effort to stimulate the economy, President George W. Bush presided over a tax plan overhaul that included changes in dividend and capital gains tax rates. The new tax plan, which was implemented in 2003, called for a 15 percent tax rate on both dividends and capital gains for investors in higher tax brackets. For lower-tax bracket investors, the tax rate on dividends and capital gains was set at 5 percent through 2007, dropping to zero in 2008.

9. **Ex-Dividend Stock Prices [LO1]** How do you think this tax law change affected ex-dividend stock prices?
10. **Stock Repurchases [LO4]** How do you think this tax law change affected the relative attractiveness of stock repurchases compared to dividend payments?

QUESTIONS AND PROBLEMS



BASIC

(Questions 1–9)

1. **Dividends and Taxes [LO2]** Gatto, Inc., has declared a \$5.85 per share dividend. Suppose capital gains are not taxed, but dividends are taxed at 15 percent. New IRS regulations require that taxes be withheld at the time the dividend is paid. The company's stock sells for \$78.35 per share, and the stock is about to go ex dividend. What do you think the ex-dividend price will be?
2. **Stock Dividends [LO3]** The owners' equity accounts for Vulcano International are shown here:

Common stock (\$.50 par value)	\$ 20,000
Capital surplus	210,000
Retained earnings	587,300
Total owners' equity	<u>\$ 817,300</u>

- a. If the company's stock currently sells for \$42 per share and a 10 percent stock dividend is declared, how many new shares will be distributed? Show how the equity accounts would change.
 - b. If the company declared a 25 percent stock dividend, how would the accounts change?
3. **Stock Splits [LO3]** For the company in Problem 2, show how the equity accounts will change if:
 - a. The company declares a 4-for-1 stock split. How many shares are outstanding now? What is the new par value per share?
 - b. The company declares a 1-for-5 reverse stock split. How many shares are outstanding now? What is the new par value per share?
4. **Stock Splits and Stock Dividends [LO3]** Stockton Mineral Operations, Inc. (SMO), currently has 540,000 shares of stock outstanding that sell for \$83 per share. Assuming no market imperfections or tax effects exist, what will the share price be after:
 - a. SMO has a 5-for-3 stock split?
 - b. SMO has a 15 percent stock dividend?
 - c. SMO has a 42.5 percent stock dividend?
 - d. SMO has a 4-for-7 reverse stock split?

Determine the new number of shares outstanding in parts (a) through (d).
5. **Regular Dividends [LO1]** The balance sheet for Quinn Corp. is shown here in market value terms. There are 12,000 shares of stock outstanding.

Market Value Balance Sheet			
Cash	\$ 49,300	Equity	\$404,300
Fixed assets	<u>355,000</u>		
Total	<u>\$404,300</u>	Total	<u>\$404,300</u>

The company has declared a dividend of \$1.45 per share. The stock goes ex dividend tomorrow. Ignoring any tax effects, what is the stock selling for today? What will it sell for tomorrow? What will the balance sheet look like after the dividends are paid?

6. **Share Repurchase [LO4]** In Problem 5, suppose the company has announced it is going to repurchase \$17,400 worth of stock. What effect will this transaction have on the equity of the firm? How many shares will be outstanding? What will the price

per share be after the repurchase? Ignoring tax effects, show how the share repurchase is effectively the same as a cash dividend.

7. **Stock Dividends [LO3]** The market value balance sheet for Murray Manufacturing is shown here. The company has declared a 25 percent stock dividend. The stock goes ex dividend tomorrow (the chronology for a stock dividend is similar to that for a cash dividend). There are 11,000 shares of stock outstanding. What will the ex-dividend price be?

Market Value Balance Sheet			
Cash	\$ 134,000	Debt	\$116,000
Fixed assets	652,000	Equity	670,000
Total	<u>\$786,000</u>	Total	<u>\$786,000</u>

8. **Stock Dividends [LO3]** The company with the common equity accounts shown here has declared a 15 percent stock dividend when the market value of its stock is \$64 per share. What effects will the distribution of the stock dividend have on the equity accounts?

Common stock (\$1 par value)	\$ 225,000
Capital surplus	535,000
Retained earnings	<u>2,968,500</u>
Total owners' equity	<u>\$3,728,500</u>

9. **Stock Splits [LO3]** In Problem 8, suppose the company instead decides on a 4-for-1 stock split. The firm's 75-cent per-share cash dividend on the new (postsplit) shares represents an increase of 10 percent over last year's dividend on the presplit stock. What effect does this have on the equity accounts? What was last year's dividend per share?
10. **Homemade Dividends [LO2]** You own 1,000 shares of stock in Avondale Corporation. You will receive a \$3.45 per share dividend in one year. In two years, the company will pay a liquidating dividend of \$62 per share. The required return on the company's stock is 15 percent. What is the current share price of your stock (ignoring taxes)? If you would rather have equal dividends in each of the next two years, show how you can accomplish this by creating homemade dividends. *Hint:* Dividends will be in the form of an annuity.
11. **Homemade Dividends [LO2]** In Problem 10, suppose you want only \$1,500 total in dividends the first year. What will your homemade dividend be in two years?

-  12. **Stock Repurchase [LO4]** Erna Corporation is evaluating an extra dividend versus a share repurchase. In either case, \$53,500 would be spent. Current earnings are \$1.79 per share, and the stock currently sells for \$64 per share. There are 9,000 shares outstanding. Ignore taxes and other imperfections in answering the first two questions.
- Evaluate the two alternatives in terms of the effect on the price per share of the stock and shareholder wealth.
 - What will be the effect on the company's EPS and PE ratio under the two different scenarios?
 - In the real world, which of these actions would you recommend? Why?

INTERMEDIATE

(Questions 10–12)

CHALLENGE

(Questions 13–16)

- 13. Expected Return, Dividends, and Taxes [LO2]** The Gecko Company and the Gordon Company are two firms that have the same business risk but different dividend policies. Gecko pays no dividend, whereas Gordon has an expected dividend yield of 2.9 percent. Suppose the capital gains tax rate is zero, whereas the income tax rate is 35 percent. Gecko has an expected earnings growth rate of 11 percent annually, and its stock price is expected to grow at this same rate. If the aftertax expected returns on the two stocks are equal (because they are in the same risk class), what is the pretax required return on Gordon's stock?
- 14. Dividends and Taxes [LO2]** As discussed in the text, in the absence of market imperfections and tax effects, we would expect the share price to decline by the amount of the dividend payment when the stock goes ex dividend. Once we consider the role of taxes, however, this is not necessarily true. One model has been proposed that incorporates tax effects into determining the ex-dividend price:⁹

$$(P_0 - P_X)/D = (1 - T_P)/(1 - T_G)$$

where P_0 is the price just before the stock goes ex, P_X is the ex-dividend share price, D is the amount of the dividend per share, T_P is the relevant marginal personal tax rate on dividends, and T_G is the effective marginal tax rate on capital gains.

- If $T_P = T_G = 0$, how much will the share price fall when the stock goes ex?
 - If $T_P = 15$ percent and $T_G = 0$, how much will the share price fall?
 - If $T_P = 15$ percent and $T_G = 30$ percent, how much will the share price fall?
 - Suppose the only owners of stock are corporations. Recall that corporations get at least a 50 percent exemption from taxation on the dividend income they receive, but they do not get such an exemption on capital gains. If the corporation's income and capital gains tax rates are both 21 percent, what does this model predict the ex-dividend share price will be?
 - What does this problem tell you about real-world tax considerations and the dividend policy of the firm?
- 15. Dividends versus Reinvestment [LO2]** National Business Machine Co. (NBM) has \$5 million of extra cash after taxes have been paid. NBM has two choices to make use of this cash. One alternative is to invest the cash in financial assets. The resulting investment income will be paid out as a special dividend at the end of three years. In this case, the firm can invest in Treasury bills yielding 2.5 percent or in 4.3 percent preferred stock. Assume IRS regulations allow the company to exclude from taxable income 50 percent of the dividends received from investing in another company's stock. Another alternative is to pay out the cash now as dividends. This would allow the shareholders to invest on their own in Treasury bills with the same yield, or in preferred stock. The corporate tax rate is 21 percent. Assume the investor has a 31 percent personal income tax rate, which is applied to interest income and preferred stock dividends. Also assume the personal dividend tax rate is 15 percent on common stock dividends. Should the cash be paid today or in three years? Which of the two options generates the highest aftertax income for the shareholders?
- 16. Dividends versus Reinvestment [LO2]** After completing its capital spending for the year, Carlson Manufacturing has \$1,000 extra cash. Carlson's managers must choose between investing the cash in Treasury bonds that yield 3 percent or paying the cash out to investors who would invest in the bonds themselves.

⁹N. Elton and M. Gruber, "Marginal Stockholder Tax Rates and the Clientele Effect," *Review of Economics and Statistics* 52 (February 1970).

- a. If the corporate tax rate is 21 percent, what personal tax rate would make the investors equally willing to receive the dividend or to let Carlson invest the money?
- b. Is the answer to part (a) reasonable? Why or why not?
- c. Suppose the only investment choice is preferred stock that yields 6 percent. The corporate dividend exclusion of 50 percent applies. What personal tax rate will make the stockholders indifferent to the outcome of Carlson's dividend decision?
- d. Is this a compelling argument for a low dividend payout ratio? Why or why not?

MINICASE

Electronic Timing, Inc.

Electronic Timing, Inc. (ETI), is a small company founded 15 years ago by electronics engineers Tom Miller and Jessica Kerr. ETI manufactures integrated circuits to capitalize on the complex mixed-signal design technology and has recently entered the market for frequency timing generators, or silicon timing devices, which provide the timing signals or “clocks” necessary to synchronize electronic systems. Its clock products originally were used in PC video graphics applications, but the market has subsequently expanded to include motherboards, PC peripheral devices, and other digital consumer electronics, such as digital television boxes and game consoles. ETI also designs and markets custom application-specific integrated circuits (ASICs) for industrial customers. The ASIC's design combines analog and digital, or mixed-signal, technology. In addition to Tom and Jessica, Nolan Pittman, who provided capital for the company, is the third primary owner. Each owns 25 percent of the 1 million shares outstanding. The company has several other individuals, including current employees, who own the remaining shares.

Recently, the company designed a new computer motherboard. The company's design is both more efficient and less expensive to manufacture, and the ETI design is expected to become standard in many personal computers. After investigating the possibility of manufacturing the new motherboard, ETI determined that the costs involved in building a new plant would be prohibitive. The owners also decided that they were unwilling to bring in another large outside owner. Instead, ETI sold the design to an outside firm. The sale of the motherboard design was completed for an aftertax payment of \$30 million.

affect the stock price? How will it affect the value of the company?

2. Jessica believes the company should use the extra cash to pay off debt and upgrade and expand its existing manufacturing capability. How would Jessica's proposals affect the company?
3. Nolan favors a share repurchase. He argues that a repurchase will increase the company's PE ratio, return on assets, and return on equity. Are his arguments correct? How will a share repurchase affect the value of the company?
4. Another option discussed by Tom, Jessica, and Nolan would be to begin a regular dividend payment to shareholders. How would you evaluate this proposal?
5. One way to value a share of stock is the dividend growth, or growing perpetuity, model. Consider the following: The dividend payout ratio is 1 minus b , where b is the “retention” or “plowback” ratio. So, the dividend next year will be the earnings next year, E_1 , times 1 minus the retention ratio. The most commonly used equation to calculate the sustainable growth rate is the return on equity times the retention ratio. Substituting these relationships into the dividend growth model, we get the following equation to calculate the price of a share of stock today:

$$P_0 = \frac{E_1(1 - b)}{R_s - \text{ROE} \times b}$$

What are the implications of this result in terms of whether the company should pay a dividend or upgrade and expand its manufacturing capability? Explain.

6. Does the question of whether the company should pay a dividend depend on whether the company is organized as a corporation or an LLC?

QUESTIONS

1. Tom believes the company should use the extra cash to pay a special one-time dividend. How will this proposal